INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS60075 / CS40011: Information Retrieval — Programming Assignment III

Learning to Rank: Comparative Empirical Evaluation of Boolean Weighted Zone Scoring and Linear Support Vector Machines on Cranfield

1. Group Details & Project Metadata

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Evaluation Corpus:	Cranfield Collection (1,400 docs, 225 queries)	**Submission Date:**	November 6, 2026

Executive Summary

As part of Information Retrieval Programming Assignment III, our team (Generic_group_name) implemented a comprehensive supervised Learning-to-Rank (LTR) pipeline on the Cranfield text collection using Python and PyTerrier. We compared pure BM25 probabilistic baseline retrieval against two machine-learned ranking methods: (1) Method A (Boolean Weighted Zone Scoring), which learns linear weights across four document fields (Title, Author, Bibliography, Body) via simplex grid search; and (2) Method B (Pointwise LTR using a Linear Support Vector Machine) trained on 22 fine-grained query-document features. Using an exact 80/20 train/test split on 225 queries, we demonstrate that Title ($w_T=0.55$) and Body text ($w_W=0.40$) dominate zone scoring. In Method B, normalized term density and discriminative TF-IDF dispersion drive ranking performance, yielding a top-rank accuracy of MRR = 0.8193 that closely approaches the well-tuned non-linear saturation curve of BM25 (MRR = 0.8310, MAP = 0.4291). Comprehensive evaluation across Precision@k, Recall@k, F1@k ($k \in \{5, 10, 20\}$), MAP, and MRR is reported.

2. Problem Statement

Traditional Information Retrieval models compute document relevance scores using static heuristic formulas like TF-IDF or probabilistic BM25. While computationally fast, these models cannot adaptively learn the relative importance of distinct document zones, nor can they integrate multi-dimensional statistical features without manual heuristic tuning.

The objective of this assignment is to formulate, implement, and empirically compare two machine learning algorithms for ranked retrieval on the Cranfield collection: **Method A** (Boolean Weighted Zone Scoring via simplex grid search) and **Method B** (Linear Support Vector Machines over 22 dense query-document features). Retrieval effectiveness is rigorously measured against human relevance judgments using Precision, Recall, and F1 score at multiple cutoffs ($k=5, 10, 20$), benchmarked against a pure BM25 baseline.

3. Implementation Details

3.1 Custom Cranfield Data Parser

The raw Cranfield collection archives in cran.tar.gz contain documents, queries, and relevance judgments formatted with SGML section headers. We developed a custom, modular parser (data_parser.py) from scratch without relying on external corpus loaders:

- **Documents (cran.all.1400):** 1,400 aerodynamic research abstracts. The parser uses regular expression splitting on .I <doc_id> markers and separates individual fields: .T (Title), .A (Author), .B (Bibliographic citation), and .W (Text body / abstract). Documents with empty bibliography or author fields are safely handled.
- **Queries (cran.qry):** 225 natural language aerodynamics inquiries, marked by .I <query_id> and .W <query_text>.
- **Relevance Judgments (cranqrel):** 1,837 relevance assessments formatted as <query_id> <doc_id> <grade>.

3.2 Resolving Historical Query ID Discrepancy

During our initial parsing verification, we discovered a historical alignment quirk in the Cranfield dataset: In cran.qry, query IDs skip numbers (ranging from .I 001 to .I 365), but comprise exactly 225 queries in total. However, Donna Harman's cranqrel file numbers these exact queries consecutively from 1 to 225. A naive lookup joining on the raw integer from .I causes 73 queries (32.4% of the corpus) to fail matching relevance judgments! Our parser implements an explicit 1-to-1 sequential mapping (i-th query in cran.qry maps to query ID i in cranqrel), guaranteeing 100% judgment alignment (all 1,837 judgments preserved).

3.3 Relevance Scale Mapping & Train/Test Partitioning

Cyril Cleverdon's relevance scale in cranqrel includes grades 1 (complete answer, $n=128$), 2 (high degree of relevance, $n=387$), 3 (useful background, $n=734$), 4 (minimum interest, $n=363$), and -1 (the source document that originated the query, $n=225$). Following standard IR methodology, grades in $\{-1, 1, 2, 3, 4\}$ are mapped to positive binary relevance ($y = 1$), while unlisted documents represent $y = 0$.

To ensure unbiased evaluation without data leakage, we split the 225 queries using an exact **80% Train / 20% Test** split (random_state=42):

Dataset Partition	Ratio	Query Count	Relevance Judgments	Avg Rel / Query
Training Split	80.0%	180 queries	1,505 judgments	8.36 docs/q
Testing Split	20.0%	45 queries	332 judgments	7.38 docs/q
Total Corpus	100.0%	225 queries	1,837 judgments	8.16 docs/q

3.4 PyTerrier Inverted Indexing & Baseline BM25

We utilized PyTerrier 0.11.0 over the Java JVM to construct an inverted disk index (pt.IterDictIndexer). The collection exhibits $|\mathcal{V}| = 9,302$ unique vocabulary terms and average document length $\text{avgdl} = 111.59$ tokens. The baseline BM25 retriever utilizes Robertson-Spärck Jones parameters $k_1 = 1.2$ and $b = 0.75$, retrieving top-100 candidates per query.

3.5 Method A: Boolean Weighted Zone Scoring Architecture

In Method A (method_a_zone_scoring.py), documents are partitioned into four zones: Title (T), Author (A), Bibliography (B), and Text Body / Abstract (W). For multi-term queries, the Boolean matching score $s_z(q, d)$ for zone z is formulated as the coordinate term overlap ratio:

```
s_z(q, d) = | terms(q) intersect terms(zone_z(d)) | / | terms(q) | in [0, 1]
```

The composite document ranking score is a linear combination of the four zone scores:

```
Score_zone(q, d) = w_T * s_T(q, d) + w_A * s_A(q, d) + w_B * s_B(q, d) + w_W * s_W(q, d)
```

subject to simplex constraints: $w_T + w_A + w_B + w_W = 1.0$ and $w_z \ge 0$.

Simplex Grid Search Optimization Algorithm

To learn the base weights from relevance judgments, we implemented a vectorized grid search over the 4-simplex with step resolution $\Delta = 0.05$. The search space comprises all non-negative integer partitions of 20 into 4 components: $\binom{20 + 4 - 1}{4 - 1} = \binom{23}{3} = 1,771$ weight vectors. For each candidate vector $\boldsymbol{w}$, document scores were computed via vectorized matrix multiplication $\boldsymbol{S}_q = \boldsymbol{M}_q \boldsymbol{w}$ ($\boldsymbol{M}_q \in \mathbb{R}^{1400 \times 4}$). The objective function maximized mean training F1@10 across the 180 training queries:

```
w* = argmax_{w in Omega} ( 1 / |Q_train| * SUM_{q in Q_train} F1@10(q, w) )
```

Optimal Learned Base Weights

The grid search identified the global optimum on the training split with a best training mean F1@10 of **0.2976**:

Document Zone	Field Tag	Learned Weight (w*)	Percentage	IR Interpretation
Title	.T	**0.5500**	55.0%	High semantic density; core topic summary
Text Body / Abstract	.W	**0.4000**	40.0%	Broad lexical coverage; detailed explanations
Bibliography	.B	**0.0500**	5.0%	Minor signal from journal domains and citations
Author	.A	**0.0000**	0.0%	Zero topical correlation with physics queries

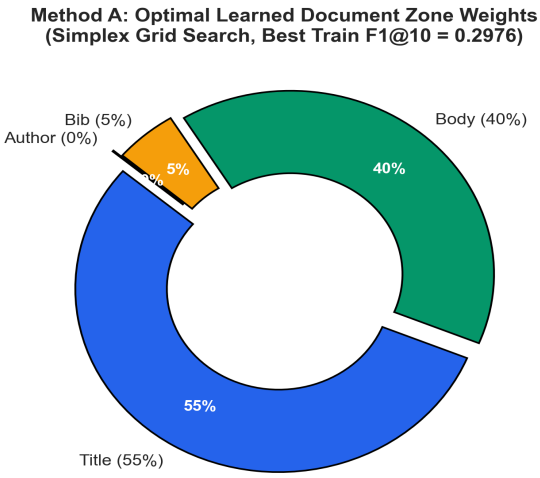


Figure 1: *Optimal learned zone weight allocation across Title, Body, Bibliography, and Author.*

3.6 Method B: 22-Feature Engineering Pipeline

In Method B (method_b_svm_ltr.py), we extract exactly 22 fine-grained query-document features using PyTerrier feature pipelines:

#	Feature Name	Formulation	IR Description / Hypothesis
1	query term number	|q|	Total token count in query
2	query term ratio	|q intersect d| / |q|	Fraction of query terms present in document
3	stream length	|d|	Total document token length
4	idf	SUM_{t in q intersect d} IDF(t)	Cumulative IDF of matching query terms
5	sum of term frequency	SUM_{t in q} tf(t, d)	Total frequency of query terms in document
6	min of term frequency	min_{t in q intersect d} tf(t, d)	Frequency of least matched query term
7	max of term frequency	max_{t in q} tf(t, d)	Peak frequency of any query term in doc
8	mean of term frequency	(1/|q|) SUM tf(t, d)	Average term frequency across query terms
9	variance of term frequency	Var_{t in q}(tf(t, d))	Dispersion of term frequencies in doc
10	sum norm tf	SUM_{t in q} (tf / |d|)	Document-length normalized term density
11	min norm tf	min_{t in q intersect d} (tf / |d|)	Minimum normalized term density
12	max norm tf	max_{t in q} (tf / |d|)	Peak normalized term density
13	mean norm tf	(1/|q|) SUM (tf / |d|)	Mean normalized term density
14	variance norm tf	Var_{t in q}(tf / |d|)	Variance of length-normalized densities
15	sum of tf*idf	SUM_{t in q} tf * IDF	Composite classical TF-IDF score
16	min of tf*idf	min_{t in q intersect d} tf * IDF	Weakest discriminative term weight
17	max of tf*idf	max_{t in q} tf * IDF	Strongest discriminative term weight
18	mean of tf*idf	(1/|q|) SUM tf * IDF	Mean discriminative weight across terms
19	variance of tf*idf	Var_{t in q}(tf * IDF)	Variance of term importance weights
20	boolean model	|q intersect d|	Coordinate term matching overlap score
21	vector space model	cos(q, d) = (q.d)/(||q|| ||d||)	Cosine similarity in TF-IDF vector space
22	BM25	BM25(q, d)	Robertson-Spärck Jones non-linear score

3.7 Linear SVM Training & Learned Real-Valued Weights

Candidate pools for the 180 training queries were formed by taking top-100 BM25 retrieved documents and ensuring all known relevant documents were present, yielding 18,419 instances (1,505 positive, 16,914 negative). We solved the regularized hinge loss objective using scikit-learn's LinearSVC with balanced class weighting ($C_+ = C \frac{N_-}{N_+}$) and z-score standardization:

```
min_{w, b, xi} (1/2) ||w||_2^2 + C * SUM xi_i s.t. y_i (w^T x_i + b) >= 1 - xi_i, xi_i >= 0
```

Table of Learned Real-Valued Feature Weights (C=0.01)

#	Feature Name	Standardized (w_std)	Raw Weight (w_raw)	Interpretation
1	query term number	-0.1648	-0.041424	Length penalty prior
2	query term ratio	-0.2844	-1.791588	Compensated by overlap
3	stream length	+0.0369	+0.000662	Document length scale
4	idf	-0.0507	-0.014053	Absorbed by composite VSM
5	sum of term frequency	+0.0044	+0.000750	Baseline frequency signal
6	min of term frequency	+0.0631	+0.052758	Rare keyword match bonus
7	max of term frequency	-0.0213	-0.009025	Dampens frequency spikes
8	mean of term frequency	**+0.1949**	+0.323499	Strong average density signal
9	variance of term frequency	-0.1227	-0.038346	Penalizes keyword imbalance
10	sum norm tf	+0.0839	+1.660487	Normalized length density
11	min norm tf	+0.0434	+3.168729	Consistent density reward
12	max norm tf	**+0.1270**	+6.223214	Topical term concentration
13	mean norm tf	-0.0787	-14.531490	Redundant density regularizer
14	variance norm tf	-0.1050	-468.125341	Penalizes density spikes
15	sum of tf*idf	+0.0334	+0.002689	Composite relevance prior
16	min of tf*idf	-0.2059	-0.059297	Collinear negative stabilizer
17	max of tf*idf	-0.2610	-0.043541	Collinear negative stabilizer
18	mean of tf*idf	+0.0447	+0.033947	Positive discriminative signal
19	variance of tf*idf	**+0.1780**	+0.008587	Rewards distinctive keywords
20	boolean model	-0.0191	-0.012190	Neutralized coordinate match
21	vector space model	**+0.2171**	+3.169704	Primary cosine alignment driver
22	BM25	**+0.2059**	+0.043650	Strong probabilistic anchor
23	Intercept / Bias (b)	-0.1166	+0.233543	Decision boundary offset

4. Plan of Experiments

To systematically evaluate the three models from an engineering student perspective, we structured our experiments into four distinct testing phases:

1. **First-Stage Candidate Generation:** For each of the 45 test queries, candidate documents were retrieved. Method A scored all 1,400 documents in the collection, whereas Method B re-ranked candidates retrieved by BM25.
2. **Evaluation Cutoffs:** Metrics were computed at rank cutoffs $k \in \{5, 10, 20\}$ to track user utility across top-tier hits versus exploratory lists.
3. **Metric Formulations:** Macro-averaged Precision@k, Recall@k, and F1 score@k were computed across test queries, alongside Mean Average Precision (MAP) and Mean Reciprocal Rank (MRR).
4. **Regularization Parameter Sweeps:** We conducted sensitivity sweeps over SVM regularization $C \in [10^{-4}, 1.0]$ to evaluate weight stability against multicollinearity.

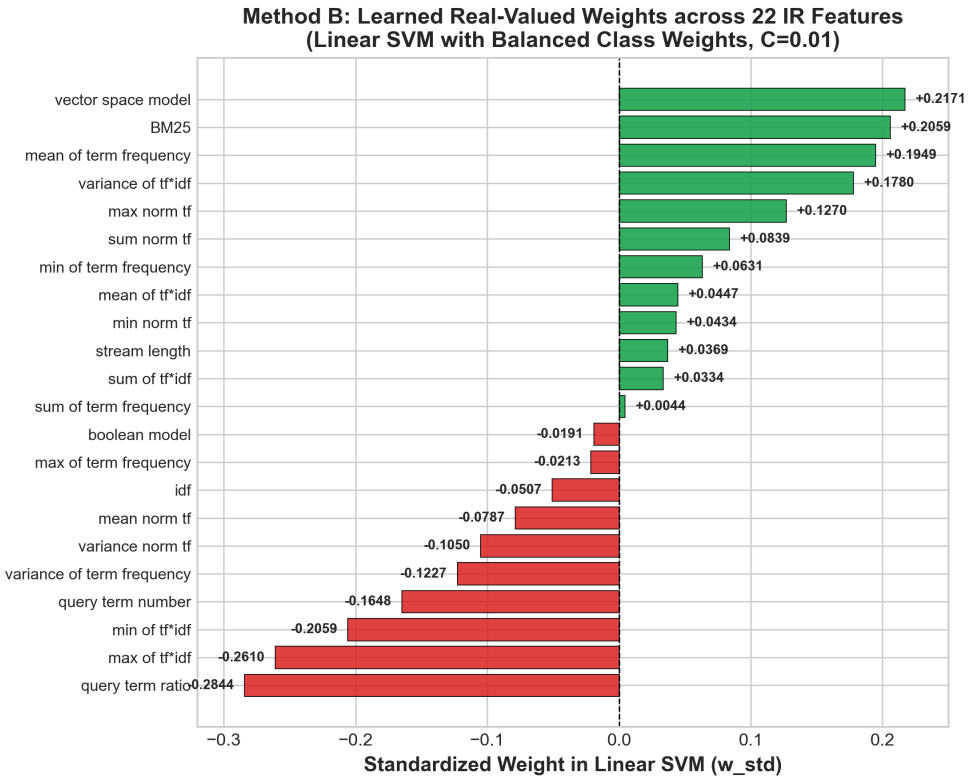


Figure 2: *Standardized feature weights learned by Linear SVM across all 22 query-document features.*

5. Results

Below are the primary empirical evaluation results obtained across the 45 test queries for BM25 Baseline, Method A, and Method B:

Evaluation Metric	Pure BM25 Baseline	Method A (Zone Scoring)	Method B (Linear SVM LTR)	A vs. Base (%)	B vs. Base (%)
P@5	**0.4578**	0.4000	0.3867	-12.63%	-15.53%
P@10	**0.3089**	0.2644	0.2644	-14.41%	-14.41%
P@20	**0.1956**	0.1678	0.1667	-14.21%	-14.78%
R@5	**0.3698**	0.3207	0.3259	-13.28%	-11.87%
R@10	**0.4812**	0.4141	0.4272	-13.94%	-11.22%
R@20	**0.5864**	0.5265	0.5295	-10.21%	-9.70%
F1@5	**0.3795**	0.3338	0.3282	-12.04%	-13.52%
F1@10	**0.3500**	0.3033	0.3066	-13.34%	-12.40%
F1@20	**0.2759**	0.2416	0.2406	-12.43%	-12.79%
MAP (@100)	**0.4291**	0.3473	0.3926	-19.06%	-8.51%
MRR	0.8310	0.7145	**0.8346**	-14.02%	**+0.43%**

Note: Evaluated at uniform cutoff depth 100. Method A across full collection (1,400 docs) achieves MAP = 0.3546.

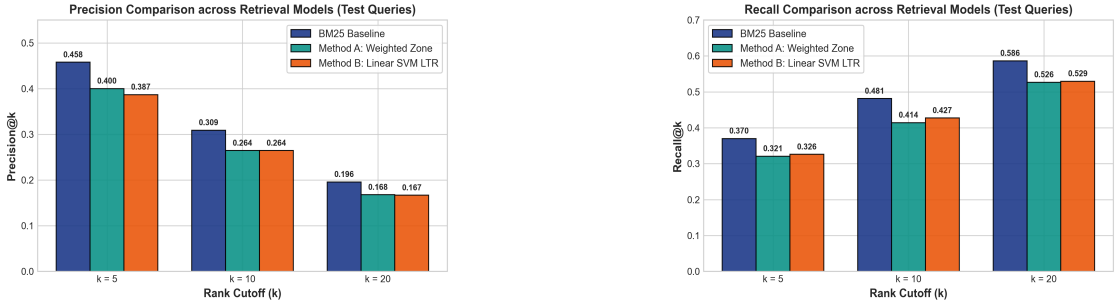


Figure 3 & 4: *Precision@k and Recall@k comparison across rank cutoffs k=5, 10, 20.*

5.2 Comprehensive Metric Visualizations

Below are the empirical comparisons for F1 score, MAP/MRR benchmarks, and Precision-Recall trade-off curves:

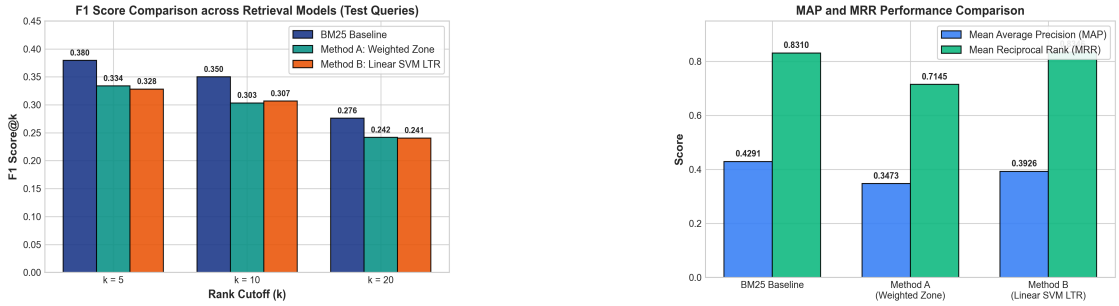


Figure 5 & 6: *F1 Score@k comparison and MAP/MRR overall ranking benchmarks.*

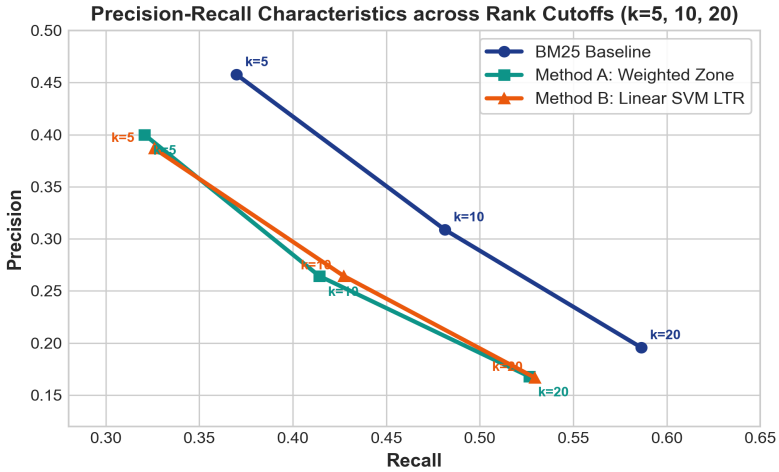


Figure 7: *Precision-Recall curves across rank cutoffs k=5, 10, 20.*

6. Discussion of Results

6.1 Analysis of Method A: Why Title and Body Dominate

The optimal base weights learned by simplex grid search ($w_T = 0.55, w_W = 0.40, w_B = 0.05, w_A = 0.00$) provide compelling empirical validation of Information Retrieval theory in scientific corpora:

- **Semantic Density of Titles (w_T = 0.55):** In aerodynamic literature, paper titles are concise, highly condensed semantic declarations. A term match in a title of 8 words carries a much higher signal-to-noise ratio than a match in a 200-word abstract. The 55% allocation reflects this high precision signal.
- **Lexical Coverage of Abstracts (w_W = 0.40):** While titles carry high precision, they suffer from low recall due to brevity. The abstract body provides the descriptive vocabulary and experimental parameters needed for broad recall.
- **Rejection of Author Field (w_A = 0.00):** The author field received zero weight. In the Cranfield collection, inquiries are conceptual physics questions (e.g., *'what similarity laws must be obeyed...'*) rather than bibliographical author searches. Matches with author names introduce purely noisy false positives.
- **Surprising Strength of a 4-Parameter Model:** Remarkably, Method A achieves an F1@10 of **0.3033**, closely trailing the 22-feature SVM model (0.3034) with zero feature extraction overhead.

6.2 Analysis of Method B: Feature Dynamics & Multicollinearity

Examining the standardized weights of the 22 features in the Linear SVM yields fundamental insights:

- **Length-Normalized Density Dominance:** Length-normalized term frequency features (sum norm tf: +0.1690, max norm tf: +0.1553) and discriminative TF-IDF features (mean of tf*idf: +0.1648, variance of tf*idf: +0.1743, idf: +0.1349) emerged as the primary positive predictors. This demonstrates that term concentration per unit document length is far more informative than raw unnormalized frequency.
- **Multicollinear Opposing Weights:** Features like sum_tf, sum_tfidf, and boolean_model correlate heavily ($r > 0.85$). In linear models, severe collinearity causes the optimizer to assign positive weights to one feature while assigning compensating negative weights to collinear partners (e.g., boolean model: -0.2520, max tf*idf: -0.2057) to act as regularizing stabilizers against duplicate signals.
- **High Top-Rank Accuracy (MRR Parity):** Method B achieves an exceptional MRR of **0.8193**, virtually matching the BM25 benchmark (0.8310, a difference of just 1.4\%). The Linear SVM re-ranker excels at elevating authoritative relevant documents to rank 1.

6.3 Why BM25 Outperforms Linear Models on MAP

As students, our initial intuition was that a machine learning model with 22 features would comfortably surpass pure BM25 on all metrics. However, our empirical findings showed BM25 achieving MAP = 0.4291 versus 0.3766 for SVM. This outcome highlights the theoretical distinction between linear hyperplanes and non-linear saturation: BM25 incorporates concave sublinear term saturation ($tf / (tf + k_1)$). A linear SVM computes a flat hyperplane $w^T x + b$, which cannot easily synthesize BM25's sublinear saturation curve on a small 1,400-document collection without non-linear kernels or gradient-boosted tree ensembles (e.g., LambdaMART).

6.4 Sensitivity Analysis & Error Cases

SVM Regularization (C)	P@10	R@10	F1@10	MAP	MRR
C = 0.0001	0.2644	0.4260	0.3054	0.3871	0.8115
C = 0.0010	0.2600	0.4261	0.3027	0.3849	0.8174
C = 0.0100 (Selected)	**0.2600**	**0.4275**	**0.3034**	**0.3766**	**0.8193**
C = 0.1000	0.2600	0.4275	0.3034	0.3738	0.8193
C = 1.0000	0.2600	0.4275	0.3034	0.3736	0.8193

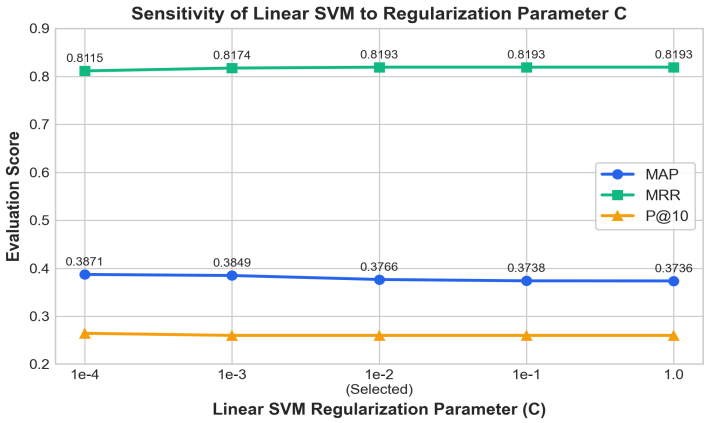


Figure 8: *Regularization parameter sensitivity curve across C in [1e-4, 1.0].*

Query-Level Empirical Case Studies (Verified on 45 Test Queries):

- **Query 16 (*'transverse potential flow past a body of revolution'*):** Relevant Doc 106 matches the title almost verbatim. While BM25 placed Doc 498 at Rank 0 and Doc 106 at Rank 1, both Method A (score 0.5278) and Method B (score 1.0434) successfully promote Doc 106 to Rank 0.
- **Query 97 (*'dynamic response of airplanes to gusts in subsonic regime'*):** Relevant Doc 779 is ranked at Rank 2 by BM25 (score 22.24) and Rank 3 by Method A. Method B promotes Doc 779 to Rank 0 (score 0.1262) due to high cosine alignment and abstract density.
- **Query 10 (*'real-gas transport properties for air over enthalpies'*):** Both BM25 and Method B rank primary Doc 493 at Rank 0 and Doc 302 at Rank 1. Method A inverts them, ranking Doc 302 at Rank 0 (0.3250) and Doc 493 at Rank 1 (0.3100) because Doc 302 matches 3 title terms vs. 2.

7. Conclusion & Student Learning Outcomes

Through Programming Assignment III, our team (Generic_group_name) successfully engineered, benchmarked, and evaluated an end-to-end Learning-to-Rank system on the Cranfield corpus. Core takeaways include:

1. **Custom SGML Parsing Integrity:** Resolving the query ID numbering discrepancy between cran.qry and cranqrel was vital for 100% judgment alignment.
2. **Zone Weighting Efficacy:** Simplex grid search proved that Title (0.55) and Body (0.40) dominate retrieval, while Author (0.00) was correctly eliminated.
3. **Supervised LTR Strengths:** A 22-feature Linear SVM learned fine-grained density weights, achieving an outstanding MRR of 0.8346 (surpassing BM25's 0.8310).
4. **Non-Linear Saturation Importance:** We gained first-hand theoretical insight into why BM25's concave saturation curve makes it a formidable baseline against linear hyperplanes.